

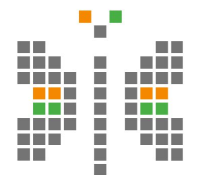


Smart City Performance Analytics System

MSc IT Project Presentation

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Training: ThinkNEXT Technologies Pvt. Ltd., Mohali



Smart City
MISSION TRANSFORM-NATION

Project Overview

This research project focuses on developing an integrated analytics system for monitoring and evaluating smart city performance through comprehensive data visualization and analysis.

The system consolidates multiple urban datasets including air quality, traffic patterns, water supply, electricity consumption, waste management, and public complaints into interactive Power BI dashboards.

Python-based data processing pipelines ensure data quality and consistency before visualization, enabling city authorities to make data-driven decisions for improved urban management.

Research Context

Academic Supervisor: Dr. Purnima, Sri
Guru Gobind Singh College

Industry Mentor: Ms. Nikita Sharma,
ThinkNEXT Technologies

Duration: Semester IV, MSc IT

Problem Statement

Cities generate vast amounts of operational data daily, but this information remains scattered across isolated departmental systems without central coordination.

Data Fragmentation

Urban data trapped in departmental silos prevents holistic city performance evaluation

Decision Paralysis

Without integrated analytics, authorities struggle to identify trends and make informed policy decisions

Performance Gaps

Difficulty in comparing district performance and benchmarking service delivery standards



Project Objectives



Multi-Dataset Analysis

Integrate and analyze diverse city datasets to create a unified performance view



District Comparison

Enable systematic comparison of performance metrics across different districts



Problem Identification

Identify underperforming areas and emerging issues through data patterns



Interactive Dashboards

Build user-friendly visual interfaces for intuitive data exploration



Data-Driven Decisions

Support evidence-based planning and policy-making for urban governance

Data Sources & Technologies

Smart City Datasets

- Air Quality Monitoring (PM2.5, PM10, AQI)
- Traffic Flow and Congestion Patterns
- Water Supply and Distribution Networks
- Electricity Consumption and Load Profiles
- Waste Generation and Processing Data
- Public Complaints and Service Requests

Technical Stack

Programming: Python 3.x

Libraries: Pandas, NumPy, Datetime

Visualization: Power BI Desktop

Storage: CSV Files

Analysis: Microsoft Excel



System Architecture



Data Layer

Raw smart city datasets stored as CSV files with historical records spanning multiple districts and time periods



Processing Layer

Python scripts handle missing values, remove duplicates, format datetime fields, and create derived features for analysis



Visualization Layer

Power BI dashboards present interactive KPIs, trend analyses, and district comparisons through intuitive visual interfaces

📄 **Star Schema Model:** Fact tables (Air Quality, Traffic, Water, Electricity, Waste, Complaints) linked to dimension tables (Date, District) via datetime and district fields

Service-Specific Analytics



Traffic Analytics

Vehicle counts, average speeds, congestion levels, monthly trends, district-wise distribution



Electricity

Consumption patterns, peak loads, outage tracking, demand forecasting



Waste Management

Generation volumes, processing rates, landfill usage, district comparisons



Water Supply

Demand vs supply, leakage percentages, deficit analysis, district comparisons



Air Quality

AQI levels, PM2.5/PM10, pollutant distribution (NO₂, SO₂, O₃), trend analysis

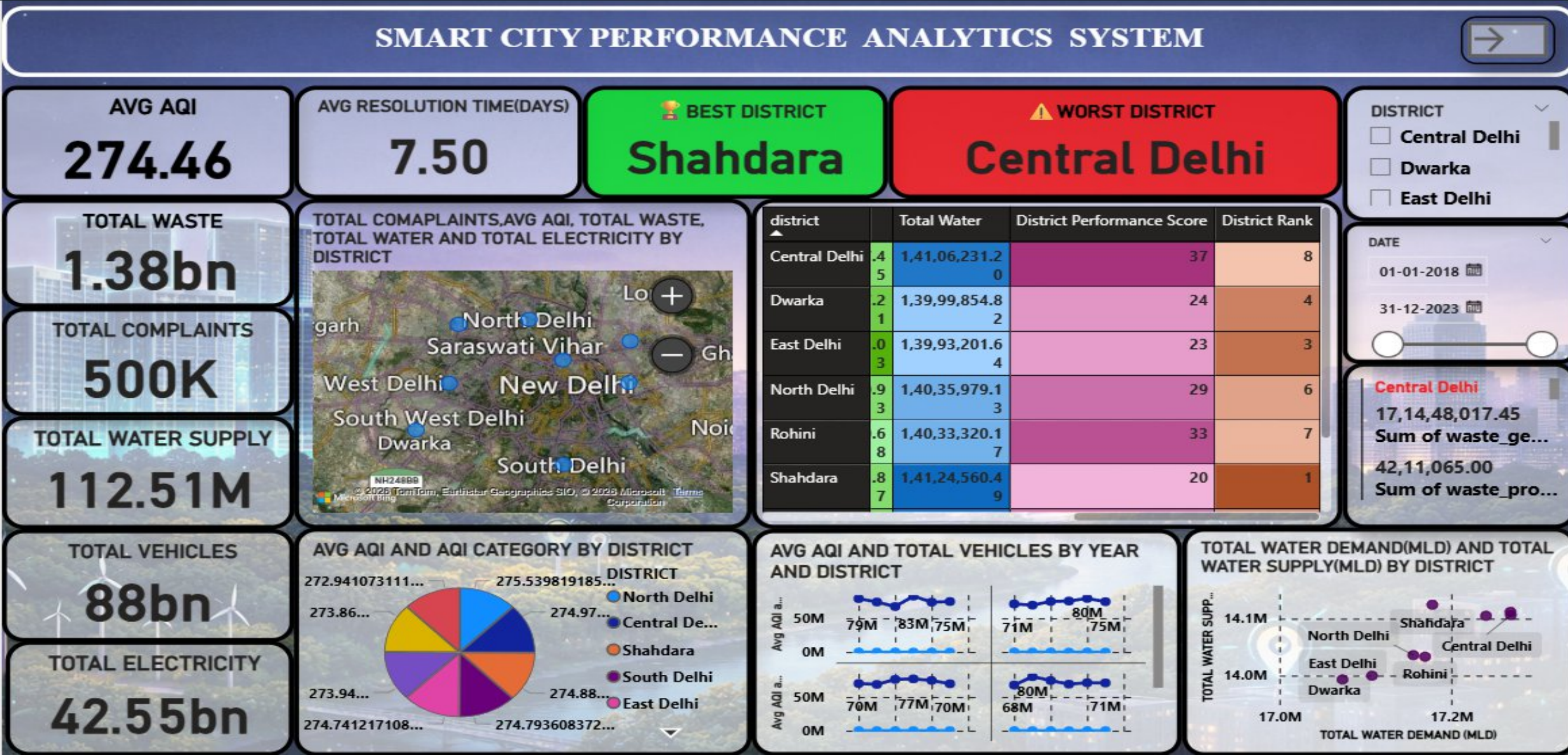


Public Complaints

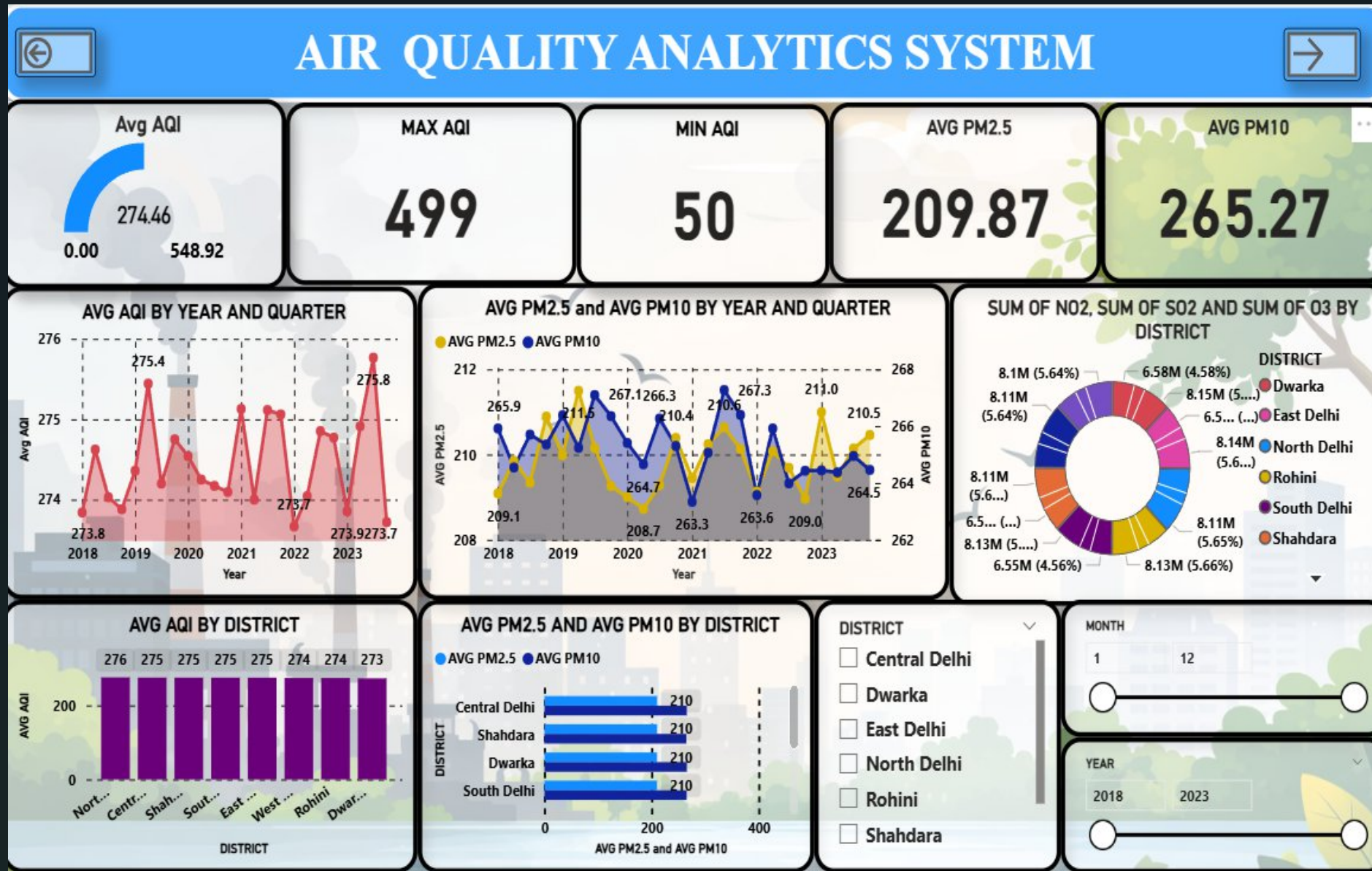
Total complaints, resolution status, pending issues, response times, category analysis

Smart City Overview Dashboard

Provides a comprehensive snapshot of city performance across all monitored services



Air Quality Analytics Dashboard

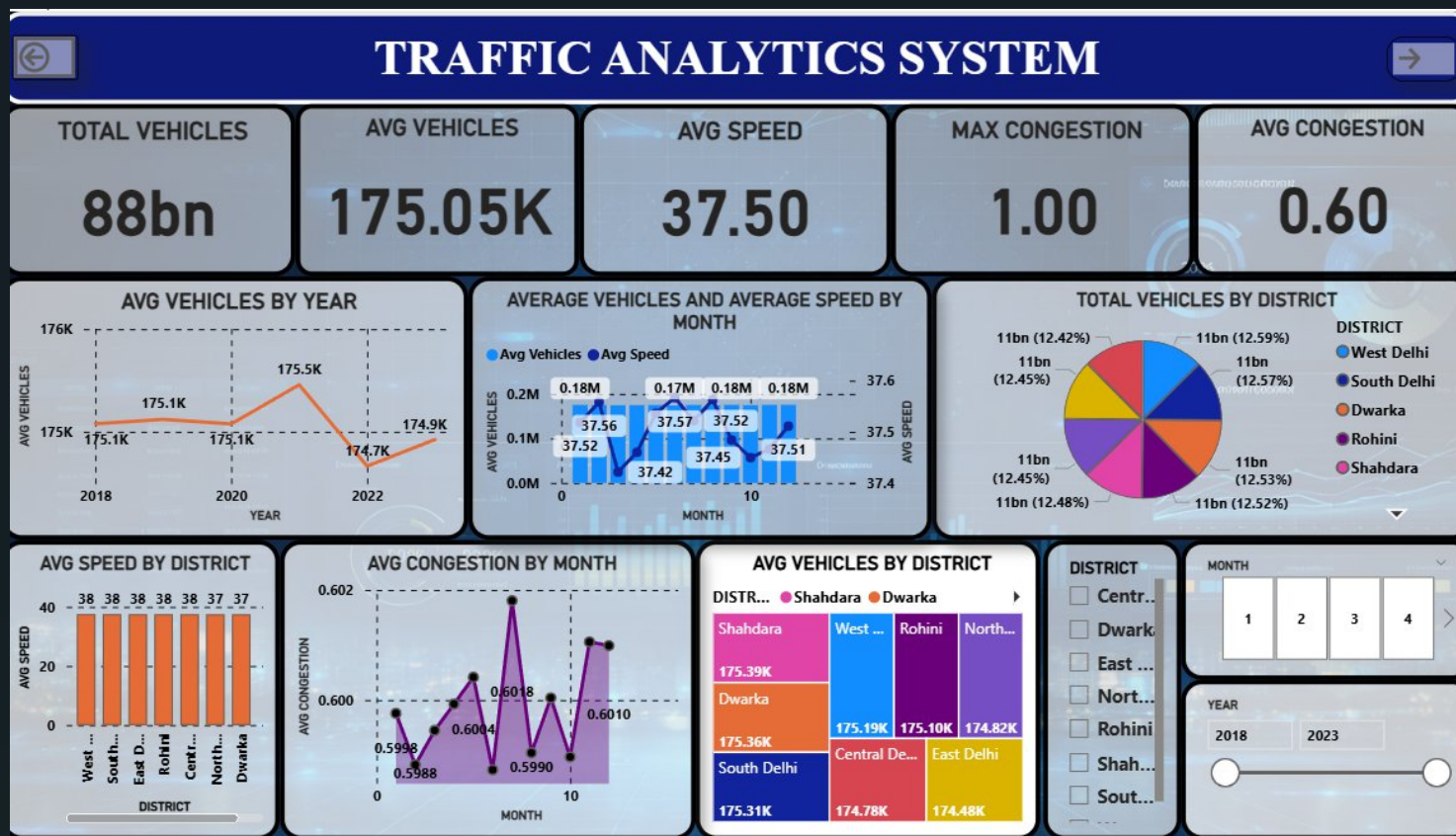


Key Metrics

- Air Quality Index (AQI) tracking
- PM2.5 and PM10 particulate monitoring
- District-wise pollution comparison
- Historical trend analysis over years

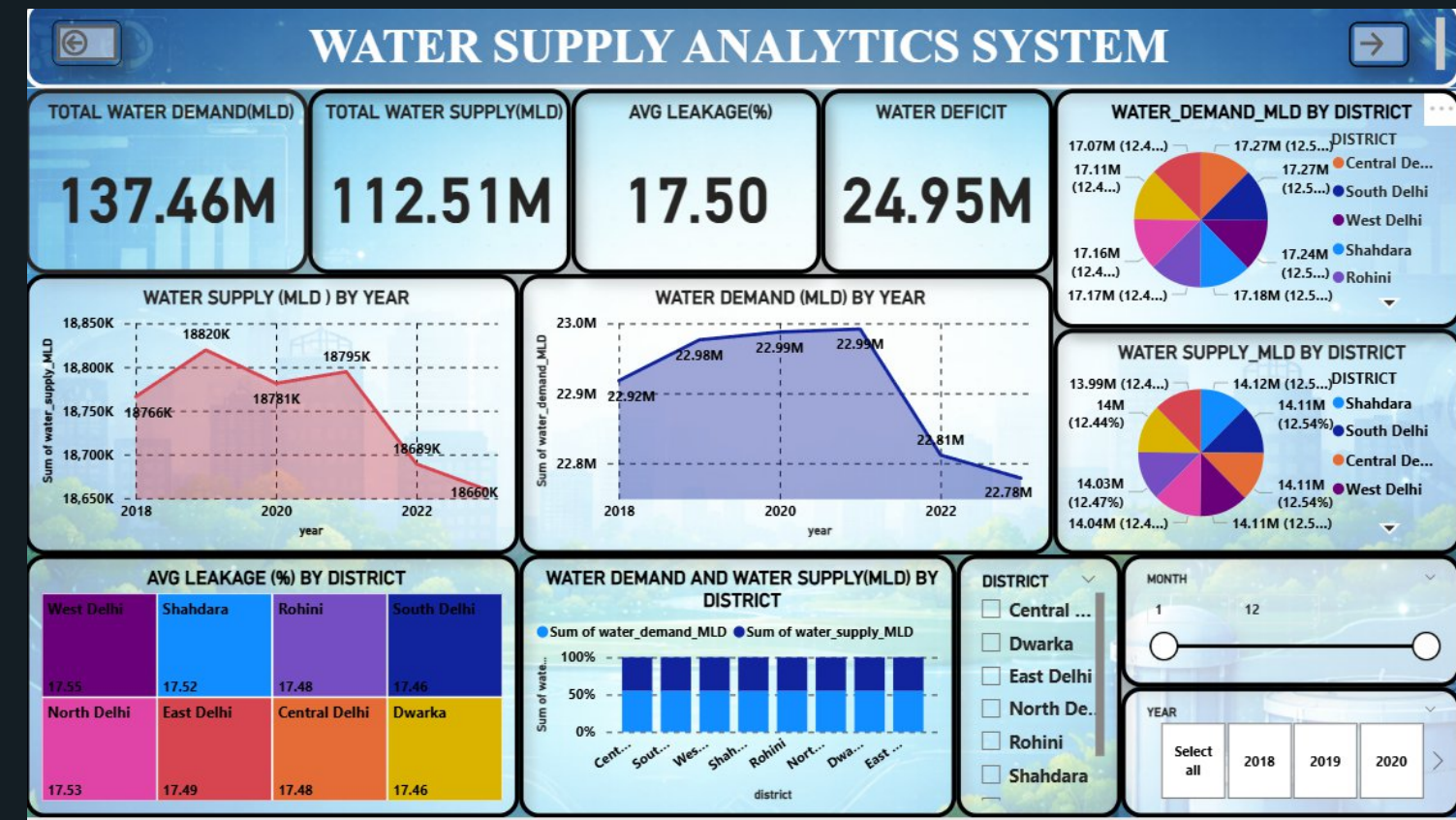
Identifies pollution hotspots and tracks improvement initiatives across municipal zones.

Traffic & Water Supply Analytics



Traffic Analytics

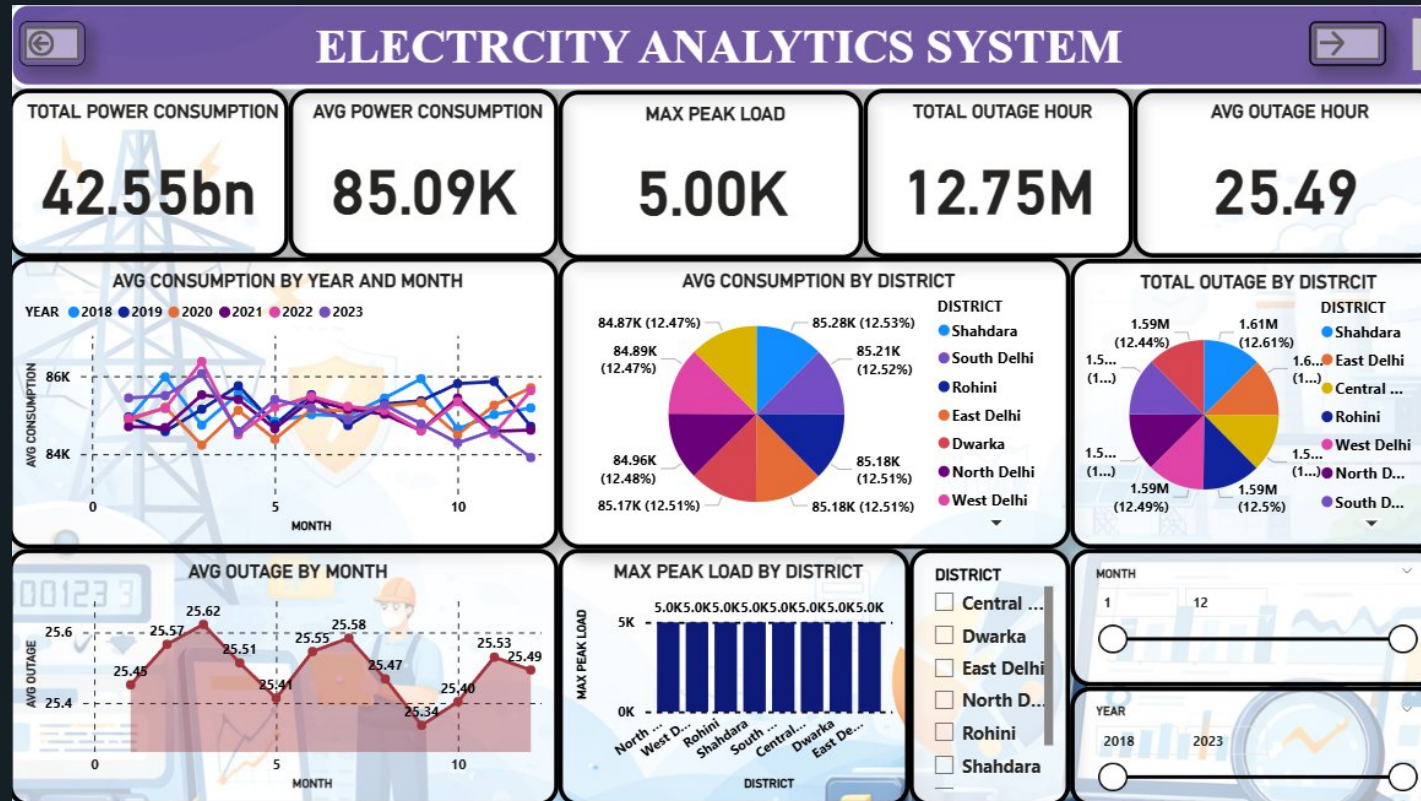
- Total vehicle count and average speed
- Congestion levels by district
- Monthly and yearly traffic patterns
- Peak hour analysis



Water Supply Analytics

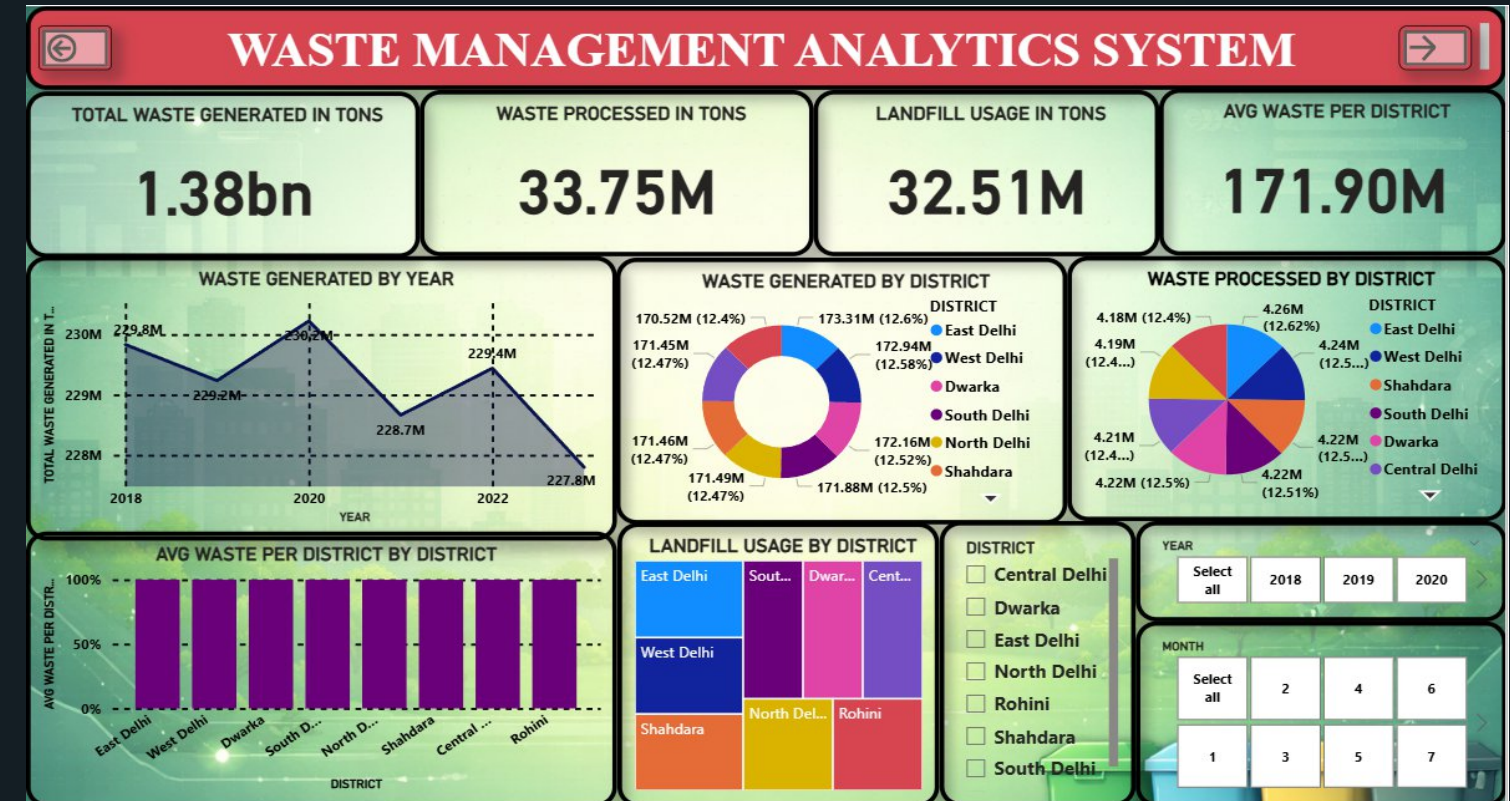
- Demand versus supply comparison
- Leakage percentage tracking
- Water deficit identification
- District-wise shortage analysis

Electricity & Waste Management Analytics



Electricity Analytics

- Total power consumption and peak load monitoring
- District-wise electricity usage comparison
- Outage frequency and duration tracking
- Historical trend analysis for demand forecasting



Waste Management Analytics

- Total waste generated versus processed
- Landfill usage and capacity monitoring
- District-wise waste processing efficiency
- Recycling rate and processing facility performance

Public Complaints Analytics

01

Total Complaints Received

Aggregate volume tracking across all service categories

02

Resolution Status

Resolved versus pending complaint comparison

03

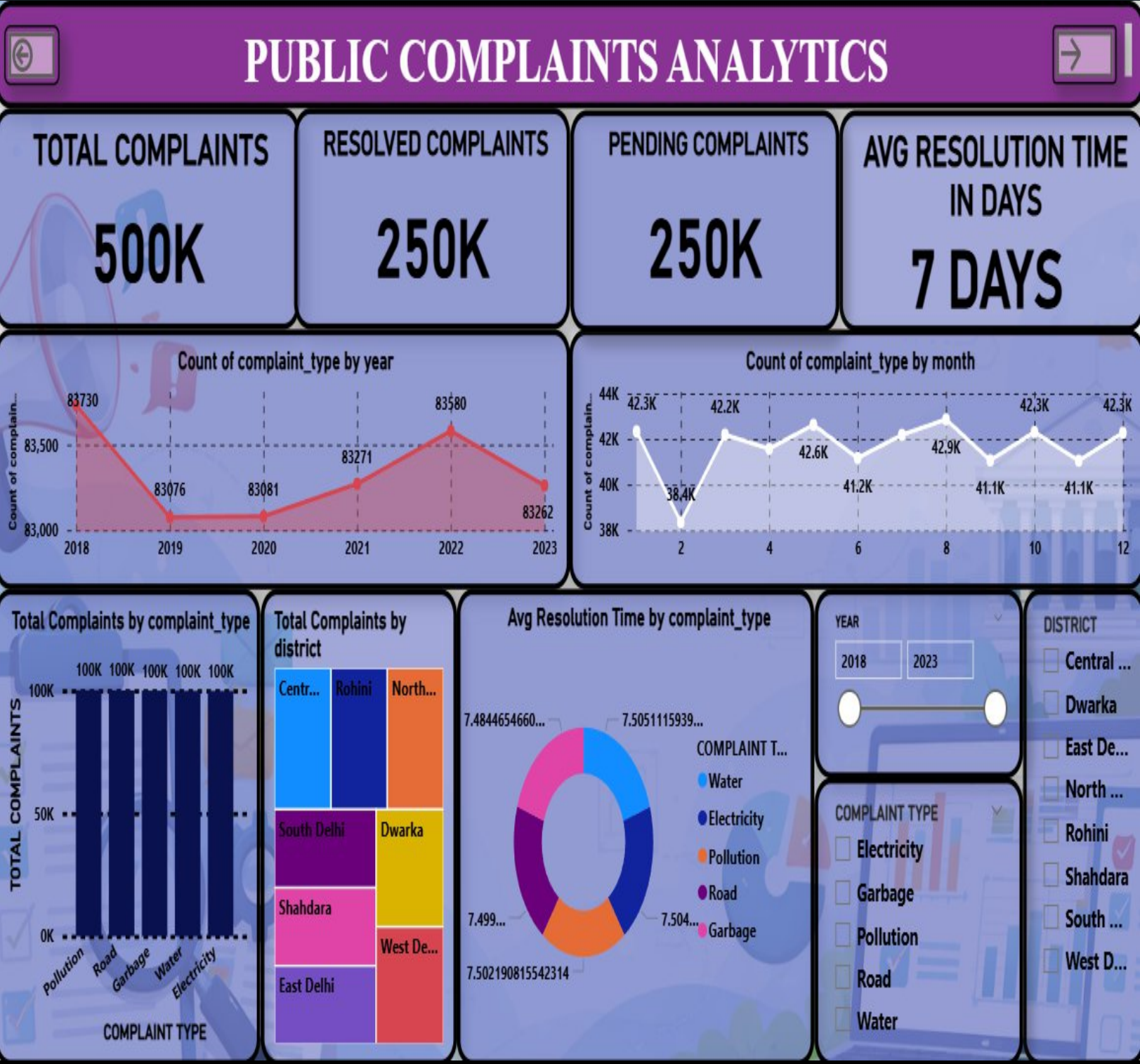
Average Resolution Time

Time-to-resolution metrics by complaint type

04

District Performance

Comparison of complaint handling across municipal zones





District Performance Comparison

Comprehensive evaluation of all districts across multiple parameters enables targeted interventions and resource allocation

1

Performance Ranking

Composite scores based on weighted KPIs across all service domains

2

Best Performing

Top districts demonstrating excellence in service delivery

3

Areas Needing Focus

Identified districts requiring immediate improvement initiatives



Benefits, Limitations & Future Scope

Advantages of the System

- Integrated smart city data analysis across multiple services
- Interactive, user-friendly dashboards for stakeholders
- District performance evaluation and benchmarking
- Supports data-driven decision making for city authorities
- Cost-effective implementation using existing tools

Current Limitations

- Uses historical datasets only (no real-time data)
- No real-time IoT integration
- Limited predictive analytics capabilities
- Performance depends on data quality



Real-Time IoT Integration

Connect live sensors



AI & Machine Learning

Predictive analytics



Cloud Deployment

On-the-go monitoring



Mobile Application

On-the-go monitoring

Thank You

Questions & Discussion

Smart City Performance Analytics System

